

IM1 Book 1 Selected Answers

IM1 Dream Team

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1. (a) $A_{ABCD} = 25$, $A_{BCEF} = 9$
(b) –
(c) –
(d) $A = 34$
(e) $l = \sqrt{34}$
(f) –
2. $l = 4\sqrt{5}$
3. –
4. $AB = \sqrt{41}$
5. $l = 5\sqrt{2}$
6. $l = \sqrt{5}$, No
7. $d = 10\sqrt{2}$
8. 12
9. $(12, 2), (2, 2)$
10. No
11. (a)

$$\begin{aligned}SR &= PQ = 13 \\RQ &= PS = 17 \\m_{SR} &= m_{PQ} = \frac{-5}{12} \\m_{RQ} &= m_{PS} = \frac{15}{8}\end{aligned}$$

- (b) It is a parallelogram but not a rectangle.
12. $(-6, 3), (10, 3)$

13. $AB = BC = \sqrt{10}$
14. (a) $(5, 0)$ – Answers may vary.
 (b) $(5, 1)$ – Answers may vary.
 (c) $x = 5$
 (d) –
15. (a) $AP = BP = 2\sqrt{5}$
 (b) $(2, 2), (3, 5), (4, 8)$
 (c) No
 (d) $y = 3(x - 1) - 1$
16. (a) $(2, 5), (3, 4)$ – Answers may vary.
 (b) $(1, 4), (2, 3)$ – Answers may vary.
17. $(1, 5)$ – Answers may vary.
 There are infinite options.
18. (a) Same size, different shape, incomparable orientations – the pieces aren't equal.
 (b) Same size, same shape, different orientations – the pieces are equal

Definition: Two figures are congruent if they have the same size and shape.

19. $y = \frac{1}{3}(x - 2) + 2$
20. $(0, 5), (3, 4), (4, 3), (5, 0), (4, -3), (3, -4), (0, -5), (-3, -4), (-4, -3), (-5, 0), (-4, 3), (-3, 4)$
21. $(2, 3), (2, -3)$
22. (a) Answers may vary: $(0, 0), (2, 5)$.
 (b) No
23. Answers may vary: $(0, 0), (4, 1), (8, 0), (4, -1)$
24. $(0, 5 + 4\sqrt{2}), (0, 5 - 4\sqrt{2})$
25. 6
26. Leg 1: $24 - 12\sqrt{2}$
 Leg 2: $24 - 12\sqrt{2}$
 Hypotenuse: $24\sqrt{2} - 24$
27. Yes
28. (a) If point P is equidistant from the coordinate axes, then point P is on the line $y = x$.

- (b) No
- (c) Answers may vary. If quadrilateral $ABCD$ is a square, then quadrilateral $ABCD$ is a rectangle.
- (d) Answers may vary. If quadrilateral $ABCD$ has four right angles, then quadrilateral $ABCD$ is a rectangle.